

Economic Dispatch Formulation

$$\min \sum_i c_i P_{G_i}$$

subject to:

$$P_{G_i}^{min} \leq P_{G_i} \leq P_{G_i}^{max}$$
$$\sum_i P_{G_i} = P_D$$

Economic Dispatch Example

What is economic dispatch for a two generator system $P_D = P_{G1} + P_{G2} = 500$ MW and

$$C_1(P_{G1}) = 1000 + 20P_{G1} + 0.01P_{G1}^2 \quad \$/hr$$

$$C_2(P_{G2}) = 400 + 15P_{G2} + 0.03P_{G2}^2 \quad \$/hr$$

Economic Dispatch Example, cont'd

We therefore need to solve three linear equations

$$20 + 0.02P_{G1} - \lambda = 0$$

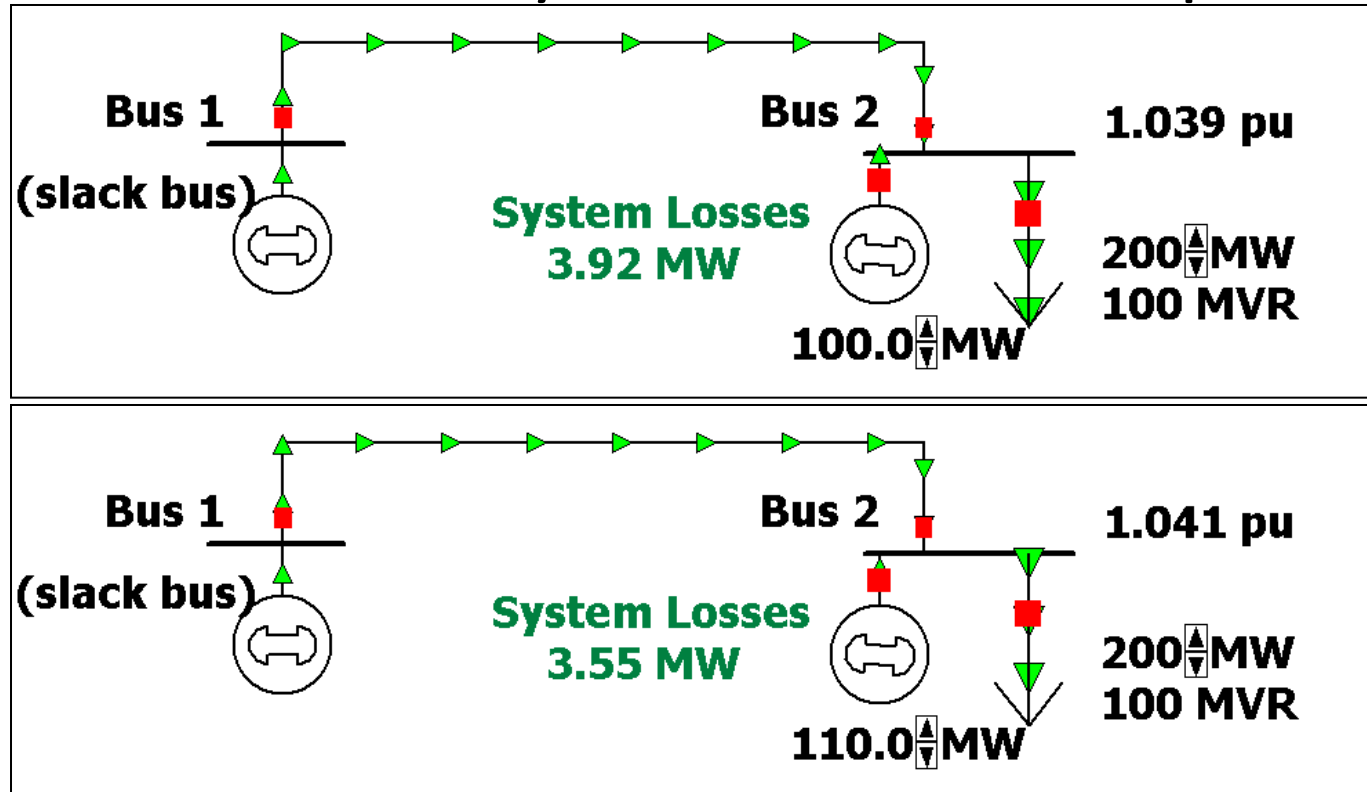
$$15 + 0.06P_{G2} - \lambda = 0$$

$$500 - P_{G1} - P_{G2} = 0$$

$$\begin{bmatrix} 0.02 & 0 & -1 \\ 0 & 0.06 & -1 \\ -1 & -1 & 0 \end{bmatrix} \begin{bmatrix} P_{G1} \\ P_{G2} \\ \lambda \end{bmatrix} = \begin{bmatrix} -20 \\ -15 \\ -500 \end{bmatrix}$$

$$\begin{bmatrix} P_{G1} \\ P_{G2} \\ \lambda \end{bmatrix} = \begin{bmatrix} 312.5 \text{ MW} \\ 187.5 \text{ MW} \\ 26.2 \text{ \$/MWh} \end{bmatrix}$$

Two Bus Penalty Factor Example



$$\frac{\partial P_L(P_G)}{\partial P_{G2}} = -0.0387$$

$$L_2 = 0.9627$$

$$\frac{\Delta P_L(P_G)}{\Delta P_{Gi}} = \frac{-0.37 \text{ MW}}{10 \text{ MW}} = -0.037$$

$$L_2 \approx 0.9643$$

AC OPF Formulation

$$\min \sum_{g \in \mathcal{G}} f(p_g^G) \quad (1)$$

subject to

$$\sum_{g \in \mathcal{G}_b} p_g^G = \sum_{d \in \mathcal{D}_b} P_d^D + \sum_{b' \in \mathcal{B}_b} p_{bb'}^L + G_b^B v_b^2 \quad (2)$$

$$\sum_{g \in \mathcal{G}_b} q_g^G = \sum_{d \in \mathcal{D}_b} Q_d^D + \sum_{b' \in \mathcal{B}_b} q_{bb'}^L - B_b^B v_b^2 \quad (3)$$

$$p_{bb'}^L = v_b^2 G_{bb} + v_b v_{b'} (G_{bb'} \cos(\theta_b - \theta_{b'}) + B_{bb'} \sin(\theta_b - \theta_{b'})) \quad (4)$$

$$q_{bb'}^L = -v_b^2 B_{bb} + v_b v_{b'} (G_{bb'} \sin(\theta_b - \theta_{b'}) - B_{bb'} \cos(\theta_b - \theta_{b'})) \quad (5)$$

$$\theta_{b_0} = 0 \quad (6)$$

$$v_b^{\text{LB}} \leq v_b \leq v_b^{\text{UB}} \quad (7)$$

$$P_g^{\text{LB}} \leq p_g \leq P_g^{\text{UB}} \quad (8)$$

$$Q_g^{\text{LB}} \leq q_g \leq Q_g^{\text{UB}} \quad (9)$$

$$p_{bb'}^L{}^2 + q_{bb'}^L{}^2 \leq (S_{bb'}^{\text{max}})^2 \quad (10)$$

DC optimal power flow

$$\min \sum_i c_i P_{G_i}$$

subject to:

$$P_{G_i}^{min} \leq P_{G_i} \leq P_{G_i}^{max}$$

$$\mathbf{B} \cdot \boldsymbol{\theta} = \mathbf{P}_G - \mathbf{P}_D$$

$$\frac{1}{x_{ij}}(\theta_i - \theta_j) \leq P_{ij,max}$$

- As an example, let's look at Transpower's DC OPF (solved every 5 minutes for real-time dispatch) [here](#)